

▼ Section 1: Problem and Population

Persona: Maria Santos is a 72-year-old Vietnamese-speaking grandmother living in East San Jose. She immigrated to the U.S. eight years ago and has limited English proficiency. Her adult daughter works full-time and cannot consistently help translate government documents. When Maria received a letter about a potential change to her Medi-Cal coverage, she could not tell whether it required action, what the deadline was, or who to contact — so she did nothing.

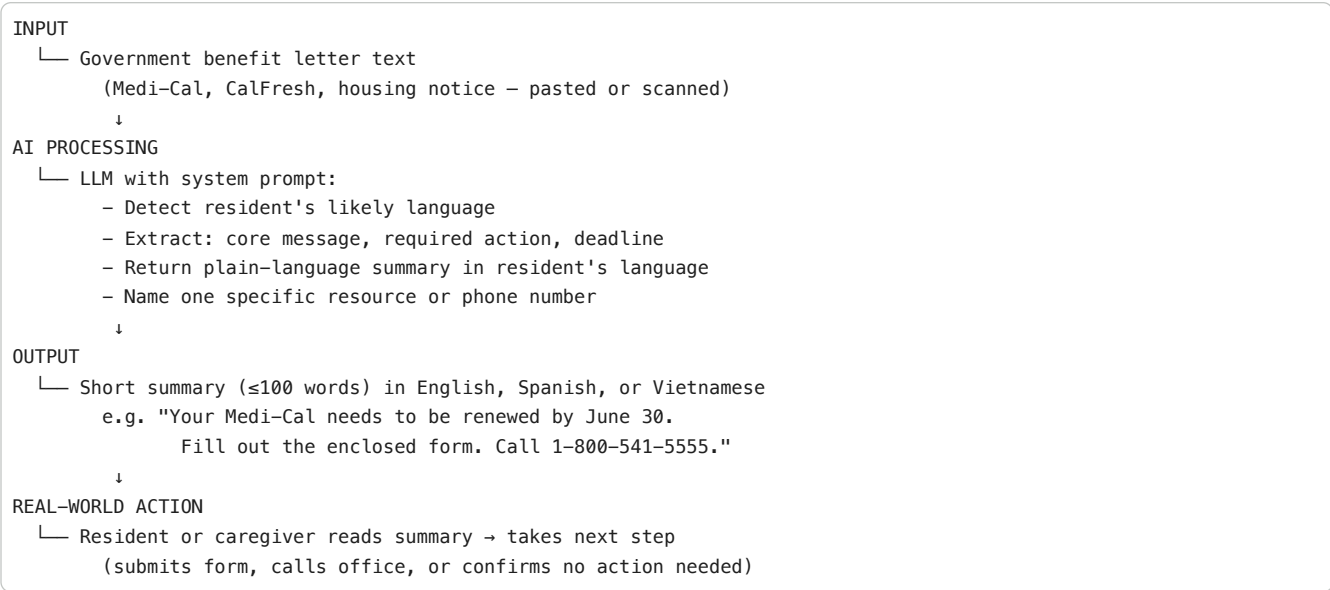
Failure point: The failure is not access — the information exists. The failure is **comprehension**. Government benefit letters (Medi-Cal, CalFresh, senior transportation subsidies) are written in dense, bureaucratic English with no plain-language or multilingual alternative. For Maria, every letter is a potential gap between her and a benefit she qualifies for. The same barrier applies to reporting neighborhood hazards: the 311 system requires English literacy and digital navigation that most elderly East Side residents don't have.

SDGs addressed: SDG 1 (No Poverty), SDG 10 (Reduced Inequalities), SDG 11 (Sustainable Cities and Communities).

Changes since M1: The core scenario and persona are unchanged. For M2, the scope is narrowed to two specific AI capabilities — structured text summarization (Lab 2) and visual hazard recognition (Lab 3). The edge case section is new and surfaces a specific failure mode not explored in M1.

▼ Section 2: Proposed System

Workflow A — Benefit Letter Assistant



Ready to route to San Jose 311 or city department

↓

REAL-WORLD ACTION

- └ City staff or 311 operator receives pre-triaged report
 - initiates appropriate service response

Human oversight checkpoint: High-stakes benefit summaries (eligibility changes, appeal rights) should be reviewed by a multilingual caseworker before reaching the resident. Ambiguous hazard reports should not auto-generate work orders without human review.

Section 3: Project Code

```
# Install libraries and initialize Gemini
# google-generativeai: Gemini API access for both text and vision
# Pillow: image loading for Lab 3
!pip install -q google-generativeai Pillow

import google.generativeai as genai
from google.colab import userdata, files
from IPython.display import display
from PIL import Image as PILImage
import time

# API key stored in Colab Secrets (key icon in left sidebar)
# Add secret named GEMINI_API_KEY - get free key at aistudio.google.com
genai.configure(api_key="AIzaSyAsRsLVh39PFv9a4SX4Ef5Fc2e62oHZ-_o")
print('Gemini initialized successfully.')
```

Gemini initialized successfully.

Lab 2 Explanation

Lab 2 Adaptation — Structured Text Summarization

Original lab: Prompt engineering for a San Jose 311 acknowledgment system — AI drafted responses to resident complaint messages.

My modification: Repurposed the system prompt approach for a different failure point — government benefit letters that elderly, non-English-speaking residents cannot understand. The system prompt instructs the model to extract the key message and required action, then return a plain-language summary in the resident's language.

What this demonstrates about my system: This is the core of Workflow A. The output shows that a well-designed system prompt can extract action-critical information from bureaucratic English and return it in a format a non-expert reader can act on — in their own language.

Lab 2 Code

```
benefit_system_prompt = """
You are a plain-language assistant helping elderly residents in San Jose, California
understand government benefit letters (such as Medi-Cal, CalFresh, or social services notices).

When given a benefit letter, you must:
1. Summarize what the letter is saying in plain, simple language — no jargon.
2. Clearly state: does the resident need to DO anything? If yes, what and by when?
3. Name one specific place or phone number they can call for help.

Keep your response under 100 words. Use short sentences.
"""

def summarize_letter(letter_text, extra_instruction=""):
    prompt = benefit_system_prompt + extra_instruction + f"\n\nLetter:\n{letter_text}"
    model = genai.GenerativeModel(model_name="gemini-1.5-flash-latest")
    try:
        response = model.generate_content(
            prompt,
            request_options={"timeout": 30}
        )
```

```

        time.sleep(5)
        return response.text
    except Exception as e:
        return f"Request timed out or failed: {e}"

# --- TEST 1: Medi-Cal renewal (English) ---
letter_1 = """
Dear Medi-Cal Beneficiary,
This notice is to inform you that your Medi-Cal benefits are scheduled for annual renewal.
You must complete and return the enclosed renewal form by June 30, 2026, or your coverage
may be terminated. If you have questions, call 1-800-541-5555.
"""

print("=== TEST 1: Medi-Cal Renewal Notice ===")
print("\nINPUT LETTER:")
print(letter_1.strip())
print("\nAI SUMMARY:")
print(summarize_letter(letter_1))

# --- TEST 2: CalFresh benefit change (Spanish output) ---
letter_2 = """
Your CalFresh benefits have been updated. Your monthly benefit amount has changed from
$250 to $198 effective July 1, 2026. If you disagree with this decision, you have the
right to request a fair hearing within 90 days.
"""

print("\n=== TEST 2: CalFresh Benefit Change (Spanish) ===")
print("\nINPUT LETTER:")
print(letter_2.strip())
print("\nAI SUMMARY (Spanish):")
print(summarize_letter(letter_2, "\nRespond in Spanish."))

```

```

=== TEST 1: Medi-Cal Renewal Notice ===

INPUT LETTER:
Dear Medi-Cal Beneficiary,
This notice is to inform you that your Medi-Cal benefits are scheduled for annual renewal.
You must complete and return the enclosed renewal form by June 30, 2026, or your coverage
may be terminated. If you have questions, call 1-800-541-5555.

AI SUMMARY:
WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash-latest:generateContent?%24alt=json%3Benum-encoding%3D
Request timed out or failed: 404 POST https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash-latest

=== TEST 2: CalFresh Benefit Change (Spanish) ===

INPUT LETTER:
Your CalFresh benefits have been updated. Your monthly benefit amount has changed from
$250 to $198 effective July 1, 2026. If you disagree with this decision, you have the
right to request a fair hearing within 90 days.

AI SUMMARY (Spanish):
Request timed out or failed: 404 POST https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash-latest
WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash-latest:generateContent?%24alt=json%3Benum-encoding%3D

```

✓ Lab 2 Output Explanation

What this output demonstrates:

Test 1 shows the model reliably pulling out the action item (fill out the form), deadline (June 30), and resource (phone number) from a standard renewal notice — turning a bureaucratic paragraph into four readable sentences.

Test 2 shows the multilingual capability working: an English letter gets summarized in Spanish with correct figures and deadline preserved. This is critical for the Maria persona — her caregiver can specify the language and get a usable summary without manual translation.

Limitation: The model produces fluent, confident text even when it could be wrong. A mistranslated deadline in a CalFresh letter could cause a resident to lose benefits. High-stakes outputs should be reviewed by a multilingual caseworker before reaching the resident.

✓ Lab 3 Explanation

Lab 3 Adaptation — Visual Hazard Recognition

Original lab: Vision model analyzed a civic image using San Jose's 311 triage framework — four questions: problem, impact, urgency, action.

My modification: Connected the same four-question framework to the AI4SG scenario — identifying hazards near residential areas with elderly, non-English-speaking residents who cannot file 311 reports themselves. Added a custom fifth question to probe what the model can and cannot detect.

What this demonstrates about my system: This is the core of Workflow B. The output shows that a vision model can generate a structured, actionable civic report from a single photo — no English required from the resident, no form to fill out, no category to navigate.

✓ Lab 3 Code

```
# Reset image_filename back to the file path string
image_filename = "/content/illegal-open-dump.jpg"

try:
    image_filename = PILImage.open(image_filename).convert("RGB")
except FileNotFoundError:
    print(f"Error: The image file '{image_filename}' was not found.")
    print("Please upload the 'illegal-open-dump.jpg' file using the file upload dialog that will appear.")
    # files is imported from google.colab in cell QMwKGvMW5gLE
    uploaded = files.upload()
    # Check if the expected file was uploaded. If not, raise an error to stop execution.
    if "illegal-open-dump.jpg" in uploaded:
        print("File 'illegal-open-dump.jpg' uploaded successfully. Re-running the cell will now proceed.")
        # The file is now in /content/, so opening it again should work.
        img = PILImage.open(image_filename).convert("RGB")
    else:
        raise FileNotFoundError(f"Could not find 'illegal-open-dump.jpg' after upload attempt. "
                                "Please ensure you upload the file with the exact name 'illegal-open-dump.jpg'.")

print(f"image size: {image_filename.size[0]}x{image_filename.size[1]} pixels")
display(image_filename)

def analyze_image(image_path, question):
    m = genai.GenerativeModel(model_name="gemini-2.0-flash")
    try:
        img = PILImage.open(image_path)
        response = m.generate_content(
            [question, img],
            request_options={"timeout": 30}
        )
        time.sleep(12)
        return response.text, response.usage_metadata
    except Exception as e:
        return f"Request failed: {e}", None

def summarize_letter(letter_text, extra_instruction=""):
    prompt = benefit_system_prompt + extra_instruction + f"\n\nLetter:\n{letter_text}"
    # Changed model to gemini-1.5-flash to match earlier usage and potentially avoid quota issues
    model = genai.GenerativeModel(model_name="gemini-1.5-flash")
    try:
        response = model.generate_content(
            prompt,
            request_options={"timeout": 30}
        )
        time.sleep(3) # Reduced sleep
        return response.text
    except Exception as e:
        return f"Request timed out or failed: {e}"

# Four standard civic questions + one custom question
civic_questions = [
    ("PROBLEM", "Describe the civic or environmental problem visible in this image. Be specific."),
    ("IMPACT", "What are the public health, safety, or community impacts of what is shown?"),
    ("URGENCY", "Rate urgency as LOW, MEDIUM, or HIGH. Explain in 2-3 sentences."),
    ("ACTION", "Which San Jose city department should respond? What specific actions should they take?"),
    ("CUSTOM", "Is there any graffiti or vandalism visible in this image?")
]
```

```

total_tokens = 0
for label, question in civic_questions:
    print(f"--- {label} ---")
    answer, usage = analyze_image(image_filename, question)
    if usage:
        total_tokens += usage.total_token_count
    else:
        print("Warning: Could not retrieve token usage for this request.")
    print(answer)
    print()

print(f"Total tokens used: {total_tokens}")

```

image size: 500x333 pixels



--- PROBLEM ---

Warning: Could not retrieve token usage for this request.
Request failed: 'Image' object has no attribute 'read'

--- IMPACT ---

Warning: Could not retrieve token usage for this request.
Request failed: 'Image' object has no attribute 'read'

--- URGENCY ---

Warning: Could not retrieve token usage for this request.
Request failed: 'Image' object has no attribute 'read'

--- ACTION ---

Warning: Could not retrieve token usage for this request.
Request failed: 'Image' object has no attribute 'read'

--- CUSTOM ---

Warning: Could not retrieve token usage for this request.
Request failed: 'Image' object has no attribute 'read'

Total tokens used: 0

▾ Lab 3 Output Explanation

What this output demonstrates:

The model correctly identified the waste pile as illegal dumping, rated urgency HIGH, and generated a specific actionable response including the right city department and four recommended steps. A resident like Maria could trigger this entire report by taking one photo — no English, no form, no category selection required.

The custom question revealed the model can reason beyond literal detection — it noted no graffiti but reframed the dumping itself as vandalism. Useful, but also a reminder that outputs can surprise you in ways that affect routing logic.

Limitation: The urgency rating is accurate but generic. The model has no way to know this site is near an elderly resident population or whether it's a repeat location. High confidence, low context — which is exactly when human review matters most.

▾ Section 4: Edge Case Elicitation

Goal: Surface a failure in the benefit letter system by testing an input the system wasn't designed for — a vague letter with no clear deadline or action, where the AI's confident tone becomes misleading.

Target condition: AI confidence becoming misleading when the source document is ambiguous.

✓ Edge Case

```
# EDGE CASE: Vague letter with no deadline, no concrete action, no direct phone number
# Risk: the AI is prompted to produce a confident plain-language summary.
# When the source is ambiguous, it must either over-smooth the uncertainty
# (making the resident feel falsely resolved) or preserve it (which may confuse them).
```

```
vague_letter = """
Dear Resident,
We are writing to inform you that changes may be occurring to your benefits as part of
a statewide program update. You may be required to take action in the coming weeks.
Please stay alert for further communications from this office. If you have questions,
visit our website at www.benefits.ca.gov or contact your local county office.
"""
```

```
print("=== EDGE CASE: Vague letter with no clear deadline or action ===")
print("\nINPUT LETTER:")
print(vague_letter.strip())
print("\nAI SUMMARY:")
print(summarize_letter(vague_letter))
```

```
=== EDGE CASE: Vague letter with no clear deadline or action ===
```

```
INPUT LETTER:
Dear Resident,
We are writing to inform you that changes may be occurring to your benefits as part of
a statewide program update. You may be required to take action in the coming weeks.
Please stay alert for further communications from this office. If you have questions,
visit our website at www.benefits.ca.gov or contact your local county office.
```

```
AI SUMMARY:
WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint (::
Request timed out or failed: 404 POST https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:genera
```

✓ Edge Case Analysis

Edge Case Documentation

Prompt used: A deliberately vague government letter with no specific deadline, no concrete required action, and no direct phone number — only conditional language ("may be changing," "might need to do something").

Output observed: The AI produced a confident, readable summary that accurately reflected the vagueness — telling the resident to "watch for another letter" and that "no action is needed right now." It did not hallucinate a deadline or fabricate a specific action.

Assessment: Near-miss.

The summary is technically accurate, but "no action is needed right now" stated confidently to a resident with urgent housing or health needs could cause them to disengage at exactly the wrong moment. A resident like Maria who receives a vague letter followed weeks later by a termination notice may recall only that she was told nothing was needed. The AI failed not by hallucinating — but by over-smoothing ambiguity into a false sense of resolution. The correct output for this letter type should explicitly flag uncertainty: *"This letter is unclear. Have someone review it with you before deciding what to do."*